



Lab 04: Cloud Monitoring Qwik Start

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Overview

Set up Cloud Monitoring for a Compute Engine VM instance running Apache. Installed monitoring and logging agents, created an uptime check, configured an alerting policy with email notifications, built a custom dashboard, and explored Cloud Logging.

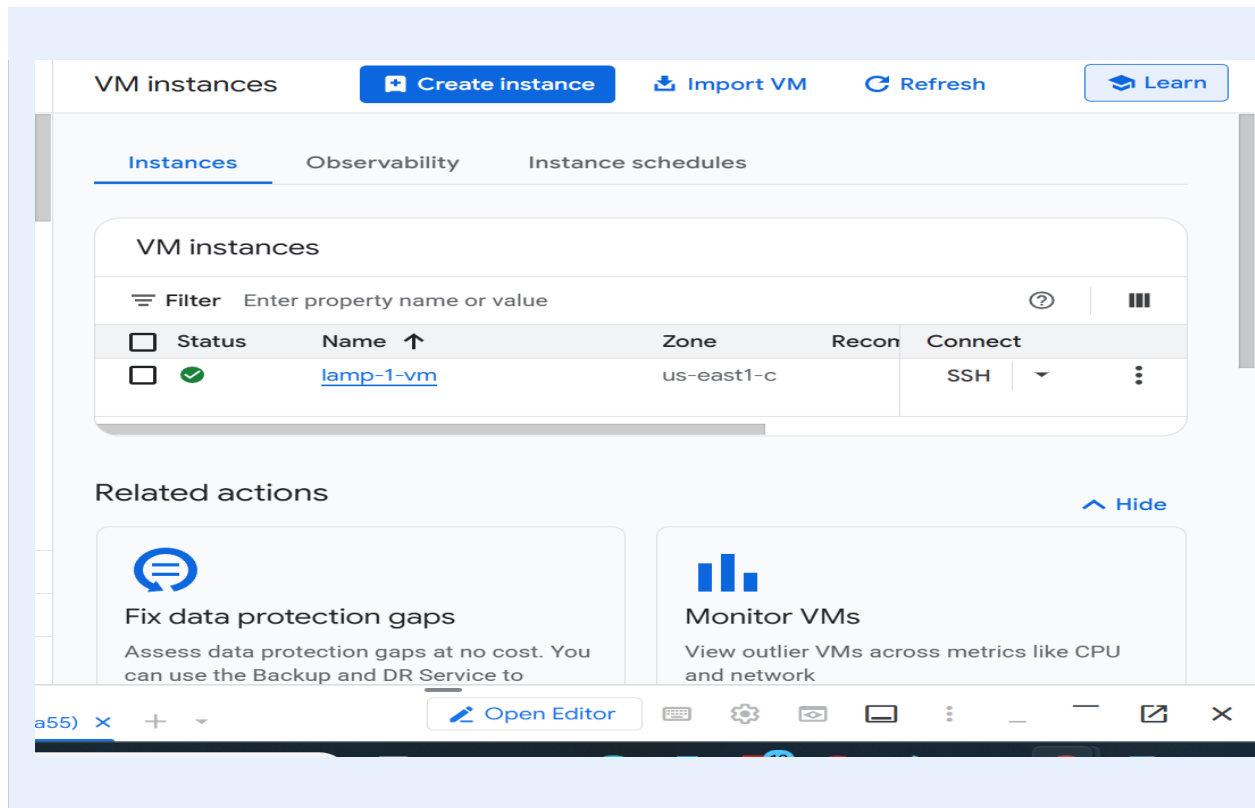
Objectives

- Create a Compute Engine VM instance with Apache installed
- Install Cloud Monitoring and Logging agents
- Create an uptime check for the VM
- Create an alerting policy for network traffic
- Build a custom monitoring dashboard with charts
- View logs in Cloud Logging
- Check uptime results and triggered alerts

Task 1 - Create a Compute Engine Instance

Created a VM instance named lamp-1-vm with Apache HTTP server and HTTP traffic enabled.

- Name: lamp-1-vm
- Series: E2, Machine type: e2-medium
- Boot disk: Debian GNU/Linux 12
- Firewall: Allow HTTP traffic



Task 2 - Install Apache and Monitoring Agents

SSH'd into the VM and installed Apache2, then installed the Cloud Ops Agent for both monitoring and logging.

```
sudo apt-get update
sudo apt-get install apache2 php7.0
sudo service apache2 restart
curl -sSO https://dl.google.com/cloudagents/add-google-cloud-ops-agent-repo.sh
sudo bash add-google-cloud-ops-agent-repo.sh --also-install
sudo systemctl status google-cloud-ops-agent"*"
```

← → ↻ ⚠ Not secure 34.139.173.118

http://45.3.63.27:60... Amazon Web Servic... SOC 101 "Essentials... app.oscarscore.com...


debian

Apache2 Debian Default Page

It works!

This is the default welcome page used to test the correct operation of the Apache2 server after installation on Debian systems. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should **replace this file** (located at `/var/www/html/index.html`) before continuing to operate your HTTP server.

If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.

Configuration Overview

Debian's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Debian tools. The configuration system is **fully documented in `/usr/share/doc/apache2/README.Debian.gz`**. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the **manual** if the `apache2-doc` package was installed on this server.

The configuration layout for an Apache2 web server installation on Debian systems is as follows:

```
/etc/apache2/
|-- apache2.conf
|   |-- ports.conf
|-- mods-enabled
|   |-- *.load
|   |-- *.conf
|-- conf-enabled
|   |-- *.conf
|-- sites-enabled
|   |-- *.conf
```

```

Reading package lists... Done
student-01-cb8aefffc48d@lamp-1-vm:~$ sudo systemctl status google-cloud-ops-agent""
● google-cloud-ops-agent.service - Google Cloud Ops Agent
   Loaded: loaded (/lib/systemd/system/google-cloud-ops-agent.service; enabled; preset: enabled)
   Active: active (exited) since Mon 2026-05-25 15:31:24 UTC; 8min ago
   Main PID: 1384 (code=exited, status=0/SUCCESS)
     CPU: 186ms

May 25 15:31:23 lamp-1-vm google_cloud_ops_agent_engine[1359]: service:
May 25 15:31:23 lamp-1-vm google_cloud_ops_agent_engine[1359]: pipelines:
May 25 15:31:23 lamp-1-vm google_cloud_ops_agent_engine[1359]:   default_pipeline:
May 25 15:31:23 lamp-1-vm google_cloud_ops_agent_engine[1359]:     receivers: [hostmetrics]
May 25 15:31:23 lamp-1-vm google_cloud_ops_agent_engine[1359]:     processors: [metrics_filter]
May 25 15:31:24 lamp-1-vm google_cloud_ops_agent_engine[1359]: 2026/05/25 15:31:24 [Ports Check] Result: PASS
May 25 15:31:24 lamp-1-vm google_cloud_ops_agent_engine[1359]: 2026/05/25 15:31:24 [Network Check] Result: PASS
May 25 15:31:24 lamp-1-vm google_cloud_ops_agent_engine[1359]: 2026/05/25 15:31:24 [API Check] Result: PASS
May 25 15:31:24 lamp-1-vm google_cloud_ops_agent_engine[1359]: 2026/05/25 15:31:24 Startup checks finished
May 25 15:31:24 lamp-1-vm systemd[1]: Finished google-cloud-ops-agent.service - Google Cloud Ops Agent.

● google-cloud-ops-agent-opentelemetry-collector.service - Google Cloud Ops Agent - Metrics Agent
   Loaded: loaded (/lib/systemd/system/google-cloud-ops-agent-opentelemetry-collector.service; static)
   Active: active (running) since Mon 2026-05-25 15:31:24 UTC; 8min ago
   Main PID: 1437 (otelopscol)
     Tasks: 8 (limit: 4616)
    Memory: 51.4M
       CPU: 5.700s
   CGroup: /system.slice/google-cloud-ops-agent-opentelemetry-collector.service
           └─1437 /opt/google-cloud-ops-agent/subagents/opentelemetry-collector/otelopscol --config=/run/goog

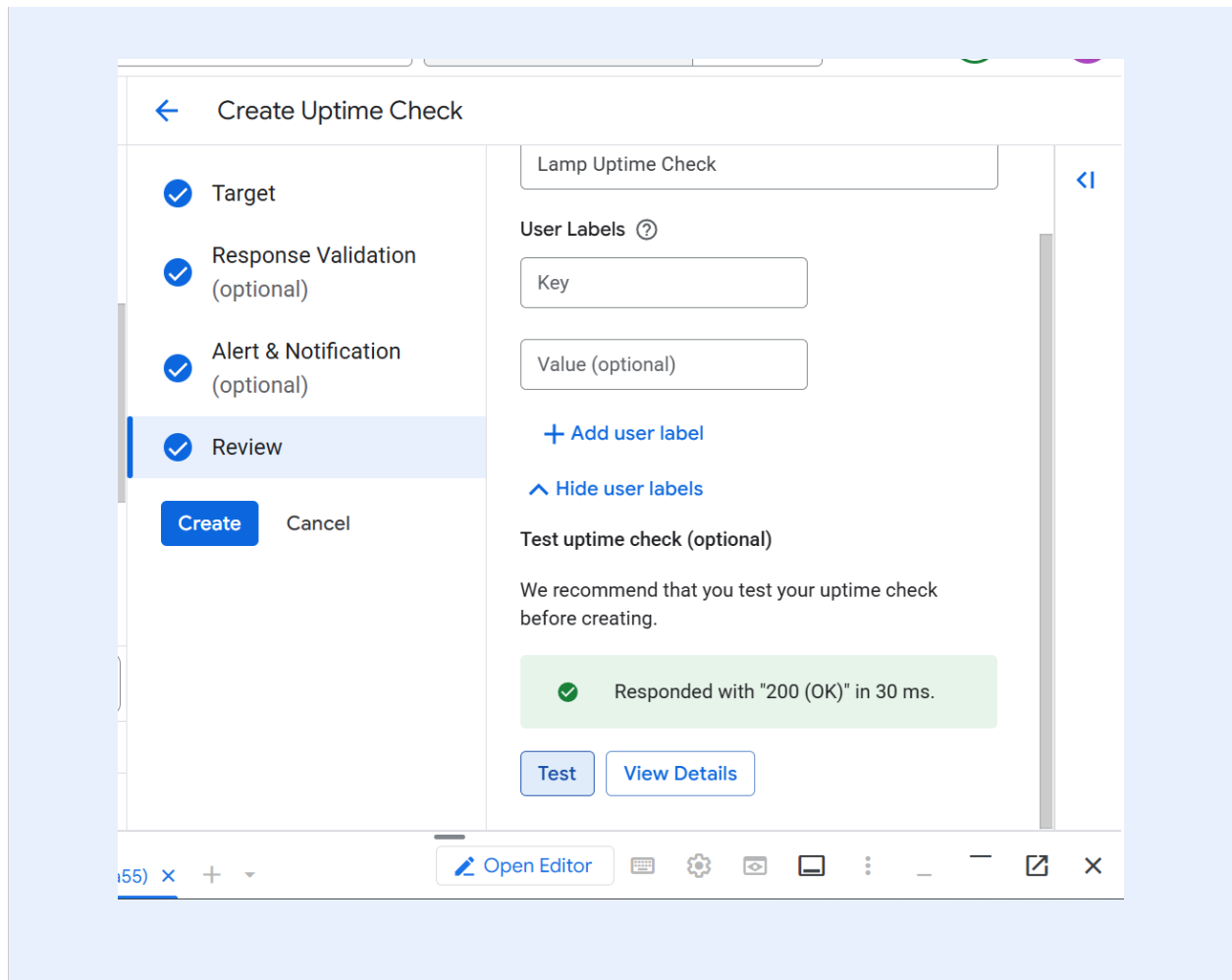
May 25 15:31:24 lamp-1-vm otelopscol[1437]: 2026-05-25T15:31:24.780Z      info      queue/persistent_queue.>
May 25 15:31:24 lamp-1-vm otelopscol[1437]: 2026-05-25T15:31:24.780Z      info      internal/resourcedetect>
May 25 15:31:24 lamp-1-vm otelopscol[1437]: 2026-05-25T15:31:24.785Z      info      internal/resourcedetect>
May 25 15:31:24 lamp-1-vm otelopscol[1437]: 2026-05-25T15:31:24.786Z      info      adapter/receiver.go:41 >
May 25 15:31:24 lamp-1-vm otelopscol[1437]: 2026-05-25T15:31:24.792Z      info      prometheusreceiver@v0.1>
May 25 15:31:24 lamp-1-vm otelopscol[1437]: 2026-05-25T15:31:24.793Z      info      targetallocator/manager>
May 25 15:31:24 lamp-1-vm otelopscol[1437]: 2026-05-25T15:31:24.793Z      info      targetallocator/manager>
May 25 15:31:24 lamp-1-vm otelopscol[1437]: 2026-05-25T15:31:24.793Z      info      service@v0.151.0/servic>
May 25 15:31:24 lamp-1-vm otelopscol[1437]: 2026-05-25T15:31:24.793Z      info      prometheusreceiver@v0.1>
May 25 15:31:24 lamp-1-vm otelopscol[1437]: 2026-05-25T15:31:24.992Z      info      fileconsumer/file.go:32>

```

Task 3 - Create an Uptime Check

Created an uptime check to verify the VM is always accessible. Configured to check every 1 minute via HTTP.

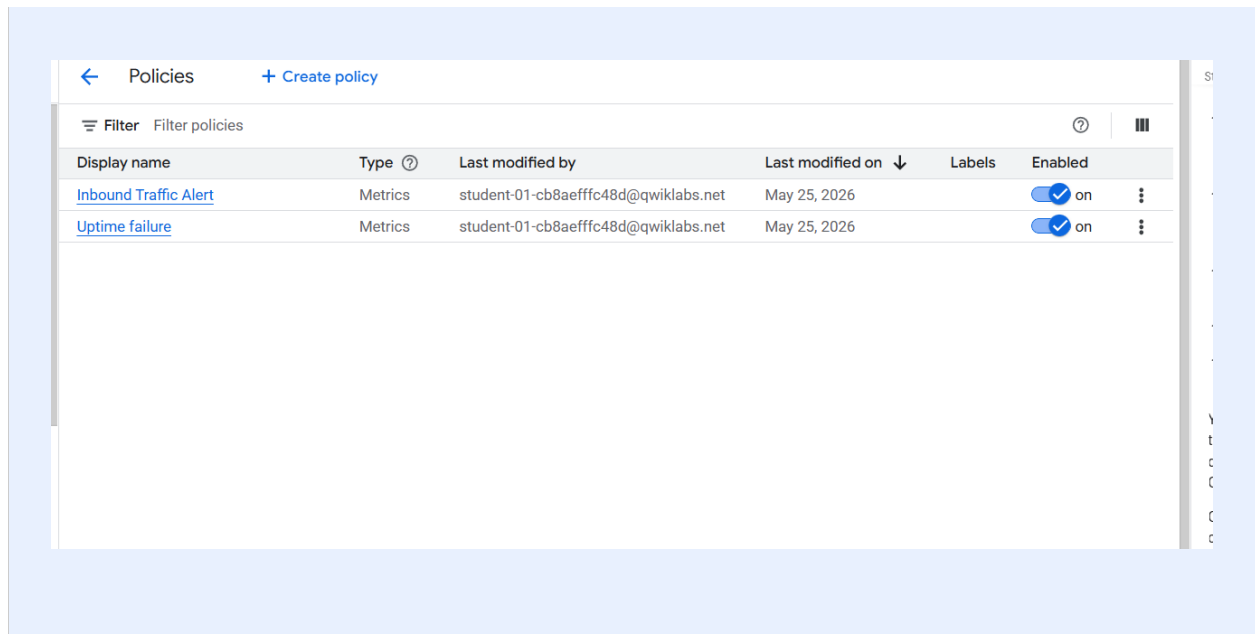
- Protocol: HTTP
- Resource type: URL
- Hostname: External IP of VM
- Check frequency: 1 minute
- Title: Lamp Uptime Check



Task 4 - Create an Alerting Policy

Created an alerting policy to send email notifications when network traffic exceeds a threshold.

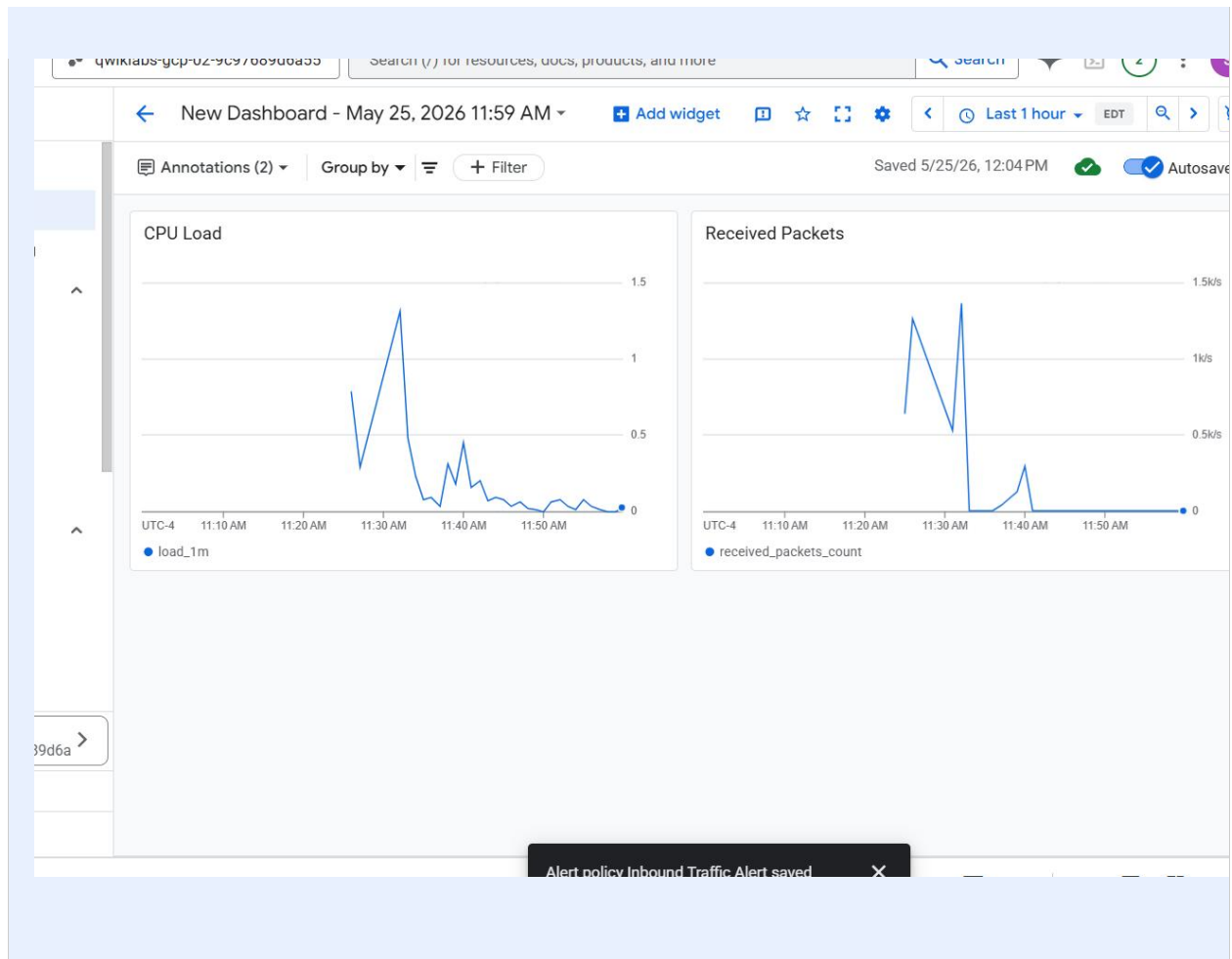
- Metric: Network traffic (agent.googleapis.com/interface/traffic)
- Threshold: Above 500
- Retest window: 1 minute
- Alert name: Inbound Traffic Alert
- Notification: Email channel



Task 5 - Create a Dashboard and Charts

Built a custom dashboard named Cloud Monitoring LAMP Qwik Start Dashboard with two line charts.

- Chart 1: CPU Load (1m) — tracks CPU usage over time
- Chart 2: Received Packets — tracks incoming network packets



Task 6 - View Logs

Explored Cloud Logging for the lamp-1-vm instance. Stopped and started the VM to observe how Cloud Logging captures VM lifecycle events in real time.

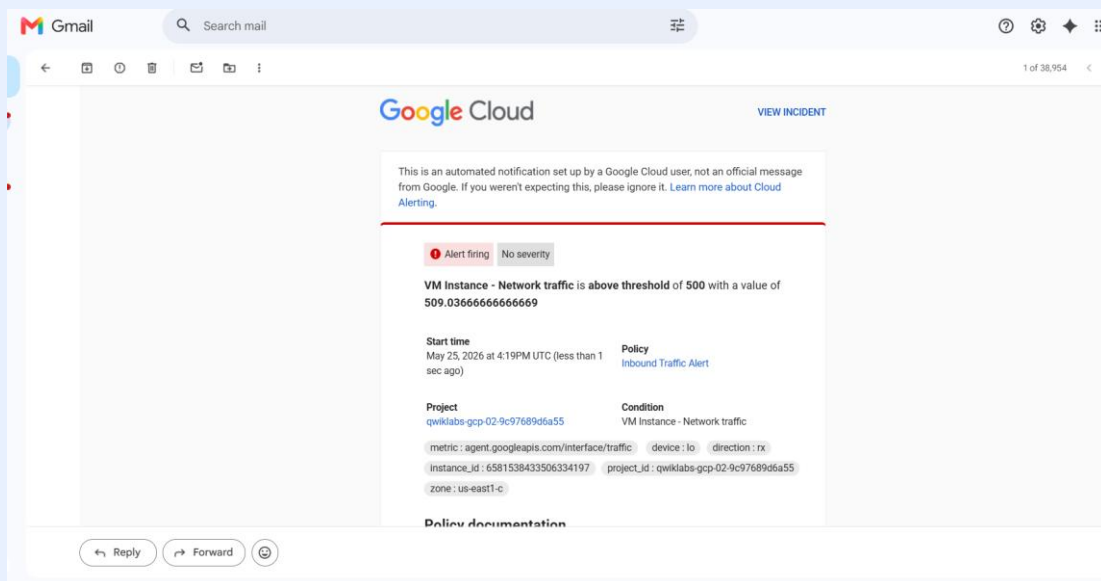
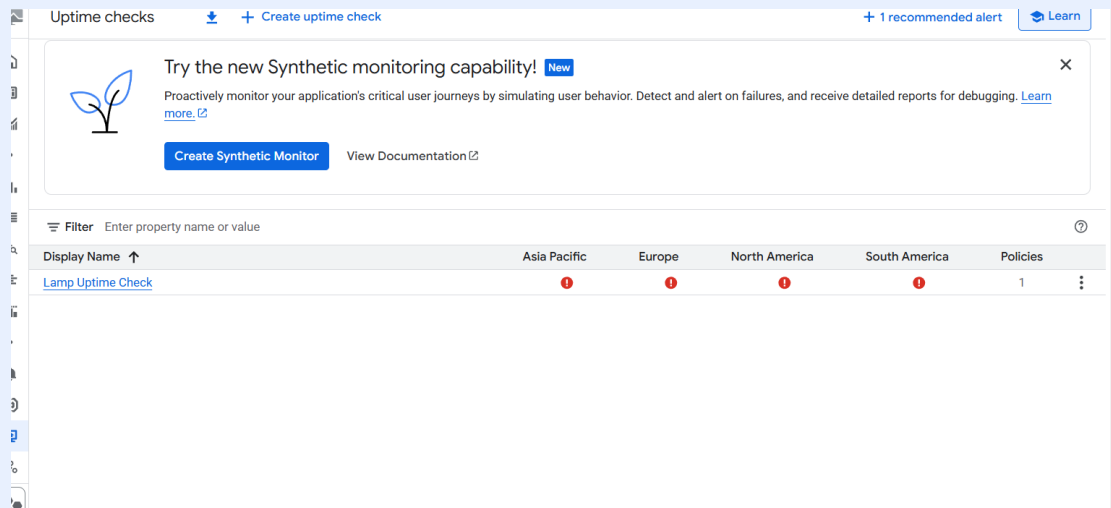
- Navigation: Logging → Logs Explorer
- Filter: VM Instance → lamp-1-vm

The screenshot displays the Google Cloud Logs Explorer interface. At the top, there are navigation links for 'Query library', 'Share link', 'Preferences', and a 'Last 1 hour' filter. Below this is a search bar with the text 'Search all fields' and a 'Run query' button. The interface is divided into several sections: 'Fields' on the left, 'Timeline' in the center, and '2,694 results' at the bottom. The 'Fields' section includes 'System Metadata' with a 'Severity' filter set to 'Default' (2,192 results) and 'Resource type' (2,694 results). The 'Timeline' section shows a horizontal timeline from 11:09:00 to 12:10:00. The '2,694 results' section displays a table of log entries. The first few entries are from 'google.monitoring.v3.UptimeCheckConfig' and 'google.monitoring.v3.UptimeCheckResult'. A notification bubble at the bottom center says 'VM instance started'.

Severity	Time	Summary
> i	2026-05-25 12:08:01.953	{ "google.monitoring.v3.UptimeCheckConfig": {...}, "google.monitoring.v3.UptimeCheckResult": {...} }
> i	2026-05-25 12:08:03.079	{ "google.monitoring.v3.UptimeCheckConfig": {...}, "google.monitoring.v3.UptimeCheckResult": {...} }
> i	2026-05-25 12:08:22.469	{ "google.monitoring.v3.UptimeCheckConfig": {...}, "google.monitoring.v3.UptimeCheckResult": {...} }
> i	2026-05-25 12:08:25.734	{ "google.monitoring.v3.UptimeCheckConfig": {...}, "google.monitoring.v3.UptimeCheckResult": {...} }
> i	2026-05-25 12:08:45.635	{ "google.monitoring.v3.UptimeCheckConfig": {...}, "google.monitoring.v3.UptimeCheckResult": {...} }
> i	2026-05-25 12:08:59.408	{ "google.monitoring.v3.UptimeCheckConfig": {...}, "google.monitoring.v3.UptimeCheckResult": {...} }
> *	2026-05-25 12:09:00.000	lamp-1-vm
> i	2026-05-25 12:09:01.920	{ "google.monitoring.v3.UptimeCheckConfig": {...}, "google.monitoring.v3.UptimeCheckResult": {...} }
> i	2026-05-25 12:09:02.522	{ "google.monitoring.v3.UptimeCheckConfig": {...}, "google.monitoring.v3.UptimeCheckResult": {...} }
> i	2026-05-25 12:09:19.848	Compute Engine start - the centos-lamp-1-vm - student-01-cb8aeffc48d@q... audit_log, method: "v1.c

Task 7 - Check Uptime and Alerts

Verified the uptime check results and confirmed that alert emails were received after the VM was restarted.



Real World Application

- DevOps teams use uptime checks to get notified immediately when a service goes down
- Alerting policies trigger on-call engineers automatically when thresholds are breached
- Custom dashboards give engineering teams a live view of application health
- Cloud Logging captures audit trails for compliance and debugging

What I Learned

- Cloud Monitoring collects metrics from VMs, apps, and third-party services
- The Cloud Ops Agent must be installed on VMs to send monitoring data
- Uptime checks verify a resource is always accessible from multiple regions
- Alerting policies can notify via email, Slack, PagerDuty, and more
- Cloud Logging and Cloud Monitoring are tightly integrated
- Stopping and starting a VM generates log entries visible in real time

Challenges & How I Solved Them

- Forgot to open the SSH terminal for the lamp-1-vm instance and attempted to install Apache2 directly from Cloud Shell instead. Cloud Shell is not connected to the VM so the install command had no effect on the instance. Fixed by clicking the SSH button next to lamp-1-vm in the Compute Engine console to open a terminal directly on the VM, then running the Apache install commands from inside that session.
- Key lesson: Cloud Shell and SSH are different. Cloud Shell runs on a separate Google-managed VM. SSH connects you directly to your specific VM instance. Always use SSH when you need to run commands on a Compute Engine instance.

Resources

GCP Cloud Monitoring Docs: <https://cloud.google.com/monitoring/docs>
Lab ID: GSP089